

W11D4 - Raccolta Informazioni (WINDOWS)

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Sommario

Oggetto del test (scope)

Risultati in breve (Firewall disattivato)

Tecnologie utilizzate

Dettaglio Risultati

⇒ OS Rilevato

Descrizione

⇒ SYN Scan

Descrizione

⇒ TCP Connect Scan

Descrizione

⇒ Services Version Detection

Descrizione

Risultati in breve (Firewall attivo)

Dettaglio Risultati

⇒ OS Rilevato

Descrizione

⇒ SYN Scan

Descrizione

⇒ TCP Connect Scan

Descrizione

⇒ Services Version Detection

Descrizione

Conclusione

Sommario

Con l'esercizio di oggi andremo a testare la macchina Windows (192.168.50.102) alla ricerca di informazioni utili per un successivo Penetration Testing.

I test effettuati comprenderanno la rilevazione del sistema operativo in uso sulla macchina target, una scansione SYN e una TCP oltre che una scansione delle versioni dei servizi eventualmente in ascolto sulle porte rilevate.

Effettueremo un primo test con il [firewall disattivato](#) e un successivo test con il [firewall attivo](#).

Le Due macchine si trovano su sottoreti diverse ma sono in grado di dialogare tra loro come si evince dagli screenshot seguenti:

- Kali Linux (attaccante)

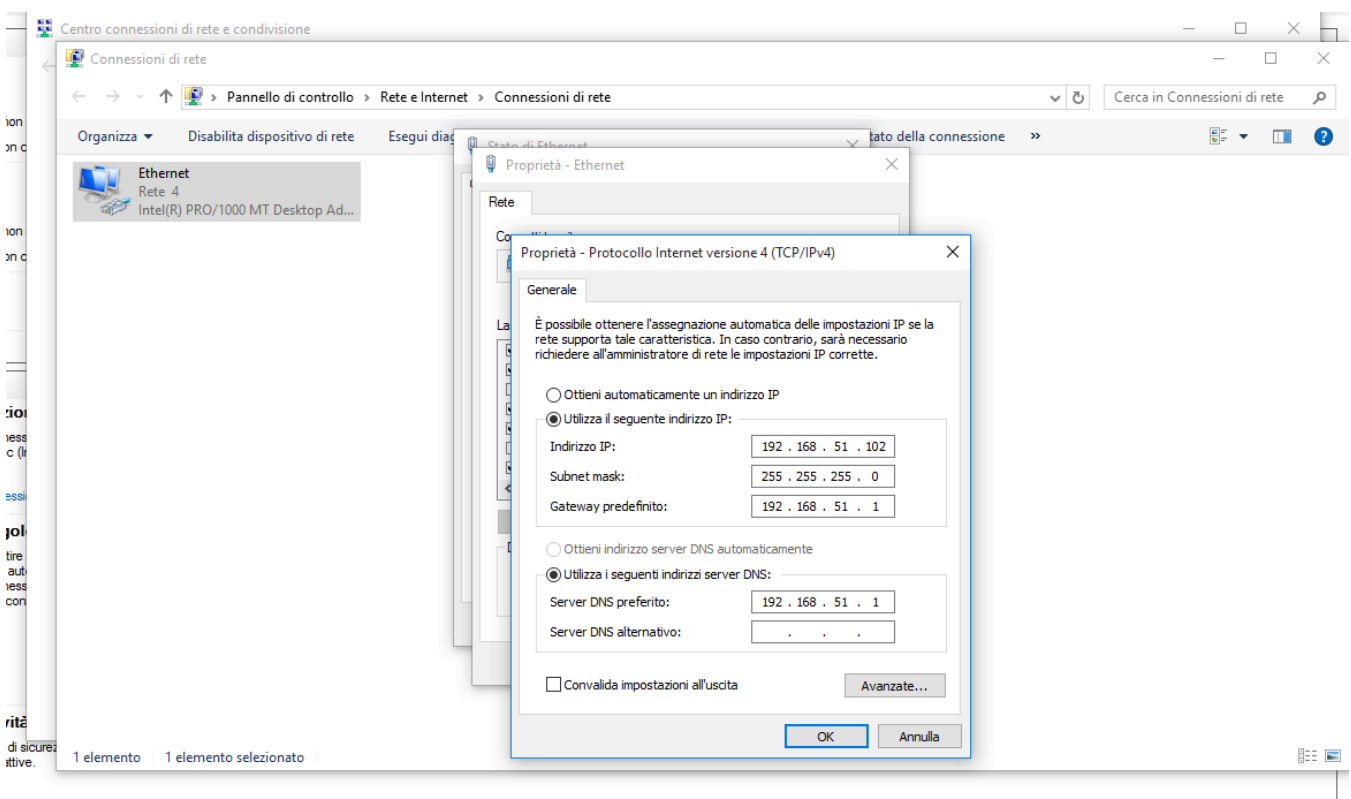
```

Session Actions Edit View Help
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:d1:f8:5d brd ff:ff:ff:ff:ff:ff
   inet 192.168.50.100/24 brd 192.168.50.255 scope global noprefixroute eth0
       valid_lft forever preferred_lft forever
   inet6 fe80::89c3:47eb:da38:43e0/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:98:9a:fd brd ff:ff:ff:ff:ff:ff

(kali@kali)-[~]
$

```

- Windows (target)



- Ping test

```

(kali@kali)-[~]
$ ping -c 4 win-nokali
PING win-nokali (192.168.51.102) 56(84) bytes of data:
64 bytes from win-nokali (192.168.51.102): icmp_seq=1 ttl=127 time=0.365 ms
64 bytes from win-nokali (192.168.51.102): icmp_seq=2 ttl=127 time=0.374 ms
64 bytes from win-nokali (192.168.51.102): icmp_seq=3 ttl=127 time=0.338 ms
64 bytes from win-nokali (192.168.51.102): icmp_seq=4 ttl=127 time=0.328 ms

— win-nokali ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3192ms
rtt min/avg/max/mdev = 0.328/0.351/0.374/0.018 ms

```

Oggetto del test (scope)

La macchina oggetto dei nostri test sarà la macchina Window 10 all'indirizzo:

- 192.168.51.102

Risultati in breve (Firewall disattivato)

#	Porte Rilevate	Stato	Servizio	Descrizione
1	7/tcp	open	echo	ECHO protocol (invia i dati ricevuti come risposta)
2	9/tcp	open	discard	DISCARD protocol (scarta i dati ricevuti)
3	13/tcp	open	daytime	daytime service, risponde con l'orario attuale
4	17/tcp	open	qotd	Quote of the day, risponde con la citazione del giorno
5	19/tcp	open	chargen	Servizio che genera e risponde con una serie di caratteri
6	80/tcp	open	http	Server Web
7	135/tcp	open	msrpc	RPC (Remote Procedure Call) protocol
8	139/tcp	open	netbios-ssn	Netbios session protocol
9	445/tcp	open	microsoft-ds	SMB (service message block) protocol
10	1801/tcp	open	msmq	Microsoft Message Queuing
11	2103/tcp	open	zephyr-clt	Zephyr Notification Service
12	2105/tcp	open	eklogin	Kerberos encrypted remote login
13	2107/tcp	open	msmq-mgmt	msmq management
14	3389/tcp	open	ms-wbt-server	Windows remote desktop and remote assistance
15	5357/tcp	open	wsdapi	Microsoft network discovery
16	5432/tcp	open	postgresql	postgresql database
17	8009/tcp	open	ajp13	Apache JServ protocol (an optimized version of the http protocol)
18	8080/tcp	open	http-proxy	HTTP_alt, porta alternativa per il webserver, spesso usata da proxy
19	8443/tcp	open	https-alt	HTTPS_alt, porta alternativa per il webserver su https

Queste sono le porte aperte e i relativi servizi in ascolto, trovati con le nostre scansioni sul target.

Tecnologie utilizzate

Per questi test abbiamo utilizzato il seguente software:

- nmap 7.95

Dettaglio Risultati

⇒ OS Rilevato

Descrizione

L'OS rilevato durante le nostre scansioni risulta essere **Windows 10**, in particolare la release **1607**.

```
(kali㉿kali)-[~]
$ nmap -O win-nokali
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 03:15 EDT
Nmap scan report for win-nokali (192.168.51.102)
Host is up (0.00026s latency).
Not shown: 981 closed tcp ports (reset)
PORT      STATE SERVICE
7/tcp     open  echo
9/tcp     open  discard
13/tcp    open  daytime
17/tcp    open  qotd
19/tcp    open  chargen
80/tcp    open  http
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
1801/tcp  open  msmq
2103/tcp  open  zephyr-clt
2105/tcp  open  eklogin
2107/tcp  open  msmq-mgmt
3389/tcp  open  ms-wbt-server
5357/tcp  open  wsdapi
5432/tcp  open  postgresql
8009/tcp  open  ajp13
8080/tcp  open  http-proxy
8443/tcp  open  https-alt
Device type: general purpose
Running: Microsoft Windows 10
OS CPE: cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 10 1607
Network Distance: 2 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.35 seconds
```

⇒ SYN Scan

Descrizione

```
(kali㉿kali)-[~]  
└─$ nmap -sS win-nokali  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 03:17 EDT  
Nmap scan report for win-nokali (192.168.51.102)  
Host is up (0.00016s latency).  
Not shown: 981 closed tcp ports (reset)  
PORT      STATE SERVICE  
7/tcp     open  echo  
9/tcp     open  discard  
13/tcp    open  daytime  
17/tcp    open  qotd  
19/tcp    open  chargen  
80/tcp    open  http  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
1801/tcp  open  msmq  
2103/tcp  open  zephyr-clt  
2105/tcp  open  eklogin  
2107/tcp  open  msmq-mgmt  
3389/tcp  open  ms-wbt-server  
5357/tcp  open  wsapi  
5432/tcp  open  postgresql  
8009/tcp  open  ajp13  
8080/tcp  open  http-proxy  
8443/tcp  open  https-alt  
  
Nmap done: 1 IP address (1 host up) scanned in 25.70 seconds
```

⇒ TCP Connect Scan

Descrizione

```
(kali㉿kali)-[~]  
$ nmap -sT win-nokali  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 03:18 EDT  
Nmap scan report for win-nokali (192.168.51.102)  
Host is up (0.00012s latency).  
Not shown: 981 closed tcp ports (conn-refused)  
PORT      STATE SERVICE  
7/tcp     open  echo  
9/tcp     open  discard  
13/tcp    open  daytime  
17/tcp    open  qotd  
19/tcp    open  chargen  
80/tcp    open  http  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
1801/tcp  open  msmq  
2103/tcp  open  zephyr-clt  
2105/tcp  open  eklogin  
2107/tcp  open  msmq-mgmt  
3389/tcp  open  ms-wbt-server  
5357/tcp  open  wsdapi  
5432/tcp  open  postgresql  
8009/tcp  open  ajp13  
8080/tcp  open  http-proxy  
8443/tcp  open  https-alt  
  
Nmap done: 1 IP address (1 host up) scanned in 10.36 seconds
```

⇒ Services Version Detection

Descrizione

```
(kali㉿kali)-[~]
$ nmap -sV win-nokali
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 03:18 EDT
Nmap scan report for win-nokali (192.168.51.102)
Host is up (0.00012s latency).
Not shown: 981 closed tcp ports (reset)
PORT      STATE SERVICE          VERSION
7/tcp     open  echo
9/tcp     open  discard?
13/tcp    open  daytime          Microsoft Windows International daytime
17/tcp    open  qotd             Windows qotd (English)
19/tcp    open  chargen
80/tcp    open  http             Microsoft IIS httpd 10.0
135/tcp   open  msrpc            Microsoft Windows RPC
139/tcp   open  netbios-ssn      Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds     Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
1801/tcp  open  msmq?
2103/tcp  open  msrpc            Microsoft Windows RPC
2105/tcp  open  msrpc            Microsoft Windows RPC
2107/tcp  open  msrpc            Microsoft Windows RPC
3389/tcp  open  ms-wbt-server    Microsoft Terminal Services
5357/tcp  open  http             Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
5432/tcp  open  postgresql?
8009/tcp  open  ajp13            Apache Jserv (Protocol v1.3)
8080/tcp  open  http             Apache Tomcat/Coyote JSP engine 1.1
8443/tcp  open  ssl/https-alt
Service Info: Host: DESKTOP-9K104BT; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 172.47 seconds
```

Tutte le scansioni hanno riportato gli stessi servizi in ascolto e le medesime porte. Chiaramente la scansione più lunga è stata l'ultima in quanto ha eseguito diversi script per estrarre le versioni dei servizi in ascolto.

Risultati in breve (Firewall attivo)

#	Porte rilevate	Stato	Servizio	Descrizione
1	80/tcp	open	http	Server Web
2	135/tcp	open	msrpc	RPC (Remote Procedure Call) protocol
3	1801/tcp	open	msmq	Microsoft Message Queuing
4	2103/tcp	open	zephyr-clt	Zephyr Notification Service
5	2105/tcp	open	eklogin	Kerberos encrypted remote login
6	2107/tcp	open	msmq-mgmt	msmq management
7	8443/tcp	open	https-alt	HTTPS_alt, porta alternativa per il webserver su https

Con il firewall attivo, il numero di porte aperte si riduce drasticamente, riducendo di conseguenza la superficie di attacco e lasciando visibili solo alcuni servizi.

Dettaglio Risultati

Come si evince dagli screenshot seguenti, per le nostre scansioni su macchina windows con firewall attivo, abbiamo dovuto usare il flag `-Pn` di nmap in modo da saltare la verifica host con ping e trattare la macchina come se fosse attiva a prescindere.

⇒ OS Rilevato

Descrizione

```
(kali@kali)-[~]
$ nmap -O win-nokali
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 03:39 EDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.47 seconds

(kali@kali)-[~]
$ nmap -Pn -O win-nokali
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 03:39 EDT
Nmap scan report for win-nokali (192.168.51.102)
Host is up (0.00029s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
135/tcp    open  msrpc
1801/tcp   open  msmq
2103/tcp   open  zephyr-clt
2105/tcp   open  eklogin
2107/tcp   open  msmq-mgmt
8443/tcp   open  https-alt
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Microsoft Windows 10 1607 (91%), Microsoft Windows 8.1 R1 (89%), Microsoft Windows Server 2008 R2 or Windows 7 SP1 (87%), Microsoft Windows Phone 7.5 or 8.0 (87%), Microsoft Windows 10 (87%), Microsoft Windows 10 1511 - 1607 (87%), Microsoft Windows Embedded Standard 7 (86%), Microsoft Windows 10 1511 (85%), Microsoft Windows 7 or Windows Server 2008 R2 (85%), Microsoft Windows Server 2016 (85%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.89 seconds
```

⇒ SYN Scan

Descrizione

```
(kali@kali)-[~]
$ nmap -sS win-nokali
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 03:41 EDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.09 seconds

(kali@kali)-[~]
$ nmap -Pn -sS win-nokali
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 03:41 EDT
Nmap scan report for win-nokali (192.168.51.102)
Host is up (0.00044s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
135/tcp    open  msrpc
1801/tcp   open  msmq
2103/tcp   open  zephyr-clt
2105/tcp   open  eklogin
2107/tcp   open  msmq-mgmt
8443/tcp   open  https-alt

Nmap done: 1 IP address (1 host up) scanned in 4.50 seconds
```

⇒ TCP Connect Scan

Descrizione

```
(kali㉿kali)-[~]
$ nmap -sT win-nokali
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 03:41 EDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.07 seconds

(kali㉿kali)-[~]
$ nmap -Pn -sT win-nokali
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 03:42 EDT
Nmap scan report for win-nokali (192.168.51.102)
Host is up (0.00073s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
135/tcp    open  msrpc
1801/tcp   open  msmq
2103/tcp   open  zephyr-clt
2105/tcp   open  eklogin
2107/tcp   open  msmq-mgmt
8443/tcp   open  https-alt

Nmap done: 1 IP address (1 host up) scanned in 8.86 seconds
```

⇒ Services Version Detection

Descrizione

```
(kali㉿kali)-[~]
$ nmap -sV win-nokali
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 03:42 EDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.55 seconds

(kali㉿kali)-[~]
$ nmap -Pn -sV win-nokali
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-23 03:42 EDT
Nmap scan report for win-nokali (192.168.51.102)
Host is up (0.00053s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  http        Microsoft IIS httpd 10.0
135/tcp    open  msrpc        Microsoft Windows RPC
1801/tcp   open  msmq?
2103/tcp   open  msrpc        Microsoft Windows RPC
2105/tcp   open  msrpc        Microsoft Windows RPC
2107/tcp   open  msrpc        Microsoft Windows RPC
8443/tcp   open  ssl/https-alt
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

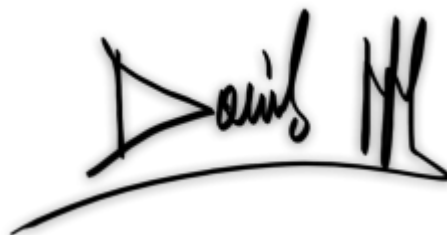
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 82.82 seconds
```

Anche in questo caso le porte rilevate dalle nostre scansioni sono le stesse in tutti i casi, con la scansione più lunga che si conferma essere quest'ultima dove effettuiamo il controllo di versione dei servizi attivi.

Conclusione

Come è facile intuire, è consigliabile mantenere attivo il firewall di Windows per ridurre la superficie vulnerabile ad un possibile attaccante.

Alla prossima.

A handwritten signature in black ink, appearing to read 'Danilo Macrì', with a long horizontal flourish underneath.